MDCNet: A Multiplatform Distributed Collaborative Network for Object Detection in Remote Sensing Imagery

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***Abstract*— With the recent development of remote sensing (RS) technology, the amount of RS platforms has witnessed a substantial increase, and the capacity of Earth observation has been greatly enhanced. The interpretation of RS images has also gradually evolved from traditional centralized ground processing to on-orbit processing. However, the traditional single-platform on-orbit processing is limited to a single source of information, which results in the underutilization of the advantages of multiplatform observation in the current RS field, and restricts the accuracy of inference tasks. To tackle the aforementioned problem, we propose a multiplatform distributed collaborative inference network, which can combine the intermediate features from multiple platforms to improve the accuracy of inference tasks. First, we proposed the collaboration map generator, which generates the collaboration map for optimal collaborator selection autonomously. Second, a spatial feature compression (SFC) module is designed to compress the interplatform transmission features, adapting spatially sparse distribution characteristics of RS objects. Finally, a feature fusion module containing spatial priors is proposed to fuse the features collected from multiple platforms to obtain more precise inference results. We conducted extensive experiments on three public datasets and verified the effectiveness of the proposed framework. On the NWPU VHR-10 dataset, for example, the proposed method improves the detection accuracy by 13.7% and 10.3% under two experimental settings compared with a single platform and compresses the intermediate data transmission between platforms by more than 80%.**

***Index Terms*— Collaborative inference, distributed processing, object detection, remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) object detection, as a significant field in the intelligent interpretation of RS images [1], [2], [3], has achieved wide applications in Earth observation tasks, including environmental monitoring [4], [5], land resource management [6], [7], natural disaster detection [8], and urban planning [9], [10], [11].

With the rapid advancement of space RS technology in recent years, the current RS platforms, in addition to acquiring and transmitting observational data, also can execute the image interpretation tasks and realize on-orbit data processing. It is particularly advantageous for some applications requiring fast response, such as disaster monitoring [12], [13]. In the study of on-orbit processing, the deployment of lightweight networks [14], [15] becomes a popular research direction, which significantly improves response efficiency. Nevertheless, this mode, primarily dependent on a single platform, cannot meet the standards of high precision in RS applications because of the low computing power and the singularity of observation.

However, the dramatic growth in the number of RS platforms offers the hardware basis for the establishment of RS platforms network, which can achieve the integration of multiple platforms for joint inference. Most importantly, through effective collaboration strategies, this mode can break the data isolation across on-orbit platforms and realize information interaction between different platforms. Eventually, in this mode, it can not only fully utilize the computational resources of each platform, but also equate to expanding the limited observation range of a single platform in time and space by merging each platform's local observation information to achieve global observation. Subsequently, the resulting cross-platform information gain aids in improving detection accuracy, as shown in Fig. 1. But, there are two major challenges to be addressed during the implementation of this mode in RS image interpretation.

The first challenge is the poor communication conditions between RS platforms, which is manifested in the following two ways. On the one hand, fully connected communication among all platforms is a common way in the process of multiplatform collaboration, where each platform interacts with information from all other platforms in collaborative inference. However, the weak communication condition in RS makes it difficult to use this all-to-all connection, which cannot ensure a stable transmission connection between platforms and results in large communication delays. On the other hand, most collaborative inference currently uses feature-level fusion [16], which can effectively reduce data redundancy while retaining the original information to the maximum extent. Although this fusion method reduces the communication cost quite

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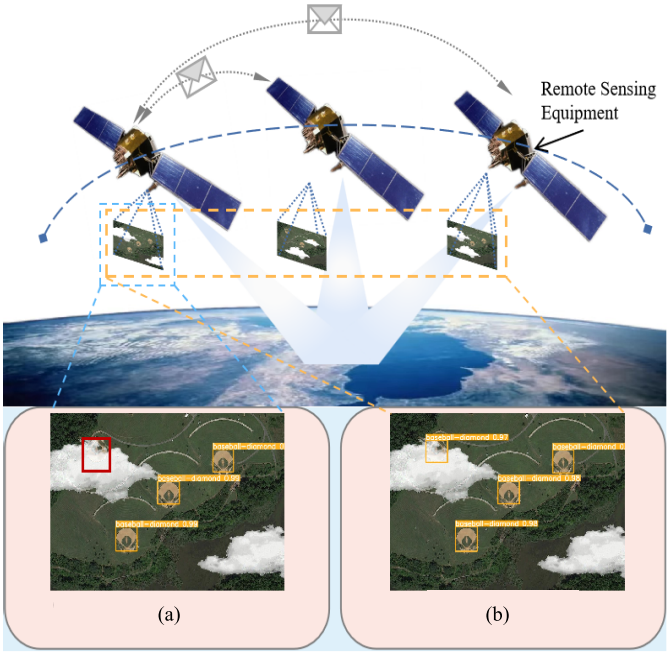


Fig. 1. Comparison between single-platform processing and multiplatform collaborative processing. (a) Results of single-platform independent inference, with red boxes indicating missed objects. (b) Results of multiplatform collaborative inference. Individual platform gets more complete results by transferring information with other platforms during the inference.

a bit compared with the original image transmission, it is still insufficient in RS applications and further exacerbates the interplatform interaction latency. The second challenge is the information interference between platforms. In the process of fusion, the information from different platforms is intersecting in both time and space dimensions, reflecting information inconsistency. During the aggregation of multiplatform information, in addition to the positive gain brought by the complementary parts, there may also be interference with each other, causing negative gain to the collaborative inference.

To address the above challenges, this article proposes an efficient multiplatform distributed collaborative inference network that aims to combine multiplatform data for high-precision on-orbit processing. Responding to the first challenge, most of the information between platforms is highly similar in the observation of the same region. Also, fully connected communication is redundant. To minimize this communication consumption, we use selective communication in multiplatform distributed collaborative network (MDCNet) without interacting with all platforms. From the information point of view, it is more reasonable to select and interact with platforms that have the most complementary information so as to maximize the information gain. Therefore, the MDCNet introduces a collaborative request matching paradigm to measure the degree of information complementarity between the requesting and the support platforms, in which the current platform will find the optimal collaborator autonomously. Furthermore, considering that in RS images, the distribution of objects tends to be spatially sparse with mostly background regions. These background features incur a large communication cost in transmission with little gain. Accordingly, we propose a spatial feature compression (SFC) module to maximize the compression of intermediate feature transmission. In the SFC module, the support platform can select and transmit the spatial region with high object perception confidence in the feature map guided by the requesting platform. These highly sparse in space but critically reliable features enable efficient and mutually beneficial collaboration. With these two approaches, MDCNet greatly eliminates redundancies and reduces communication costs in RS. Finally, toward the second challenge, we design a spatially confident feature fusion (SCFF) module. For a particular platform, the information from other platforms may interfere and detract from the results with high enough confidence. Therefore, the fusion module locally fuses features in areas with weak spatial confidence by using the collected collaboration messages from others and retaining features in areas with high confidence.

The main contributions of this article are summarized as follows.

1) This article proposes the MDCNet. Through the selective communication strategy for multiplatform feature interaction, our method improves the accuracy of object detection and optimizes the inference efficiency by breaking through the limitations of single-platform observation range and precision.
2) We present an efficient collaborative information compression strategy called SFC module, which adapts the characteristics of RS images with large-scale variations and sparse distribution of objects. Also, it compresses the amount of data transferred between platforms to less than 20% of the original image.
3) An SCFF module is presented to fuse the intermediate features, which achieves effectively mines complementary information under the guidance of a prior information provided by each platform to obtain more accurate and reliable fusion results and improve the final detection accuracy.

II. Related Work

In this section, we will introduce some related work pertaining to our proposed method. Specifically, it can be divided into the following two aspects: RS object detection and the investigation of multiagent collaborative inference.

A. RS Object Detection

Object detection is a fundamental research topic in the domain of RS image interpretation. Compared with natural images, RS images are characterized by a wider observation range and a more intricate background, and RS objects are often variable in scale and unevenly distributed. Traditional approaches usually depend on handcraft feature to locate objects, such as shape and texture [17], and scale-invariant features [18]. With the fast growth of deep learning, researchers are steadily developing novel object detection algorithms based on convolutional neural networks (CNNs). Due to their strong capability of feature extraction, they can acquire more robust object features, include rich high-level semantic information, and significantly improve detection accuracy. In recent years, many researchers have conducted extensive research on RS object detection task based on the characteristics above. There are two popular types of object

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detection methods: single-stage and two-stage approaches. The representative two-stage approaches are RCNN [19] series, including the derived Faster RCNN [20], Mask RCNN [21], and Cascade RCNN [22]. In the two-stage approach, the region of interest (ROI) needs to be preselected, which results in numerous repetitive region proposals computation, and then classified and regressed. Although this method can obtain considerable detection accuracy, the speed and efficiency of inference are limited. Classical single-stage approaches include single-shot multibox detector (SSD) [23], you only look once (YOLO) series [24], [25], [26], and RetinaNet [27]. These approaches simultaneously accomplish the classification and location regression, which achieves a great improvement in inference efficiency.

A group of researchers have introduced the above methods into RS image interpretation and adapted them with RS characteristics, producing some excellent work. You only look twice (YOLT) proposed by Etten [28] employed and enhanced the YOLO detector according to characteristics of tiny objects in wide RS images, which achieved a great balance between accuracy and inference speed. Su et al. [29] modified the feature extraction network of YOLO and added the rotation detection function, which is applied to the fast detection of ship in large RS images. Considering the symmetrical shape of RS objects, Shi et al. [30] introduced a channel attention module (CAM) to be integrated with the feature pyramid network (FPN) structure, which improved detection effect by emphasizing the central feature of the objects. By mining the contextual relationships between complex backgrounds and objects, Liu et al. [31] designed a mechanism named enhanced effective channel attention (EECA) to enhance the feature extraction capability and utilized deep-level semantic information by adaptive FPN (AFPN). Zhang et al. [32] utilized assisted super-resolution learning to integrate multimodal data and achieve more accurate and timely detection of small, multiscale objects in RS images. Zhang et al. [33] introduced a coarse-to-fine object detection framework, targeting both feature and sample, respectively. On the one hand, the feature representation extracted by the backbone is refined and enhanced, while, on the other hand, stronger samples are screened for training, resulting in better detection accuracy.

B. Multiagent Collaborative Inference

An agent refers to an intelligent individuals that have certain computational resources and can independently perform visual tasks. Currently, significant advancements have been achieved in deploying deep neural networks on a single agent to perform some scene interpretation tasks, such as drone tracking [34], [35], [36], [37], robotic motion [38], [39], and self-driving vehicles [40], [41], [42]. Meanwhile, with the advancement of productivity and technology, the number of intelligent agents in the world's scenarios has grown dramatically. Many researchers started to explore the method of multiagent collaboration to perceive a scenario from numerous perspectives.

In natural scenarios, the collaboration of multiple agents has already made good progress in several specific applications. Some researchers are primarily concerned with the formulation of communication strategies [43], [44], [45], [46], [47] in multiagent systems, which is a popular direction. Vertex attention interaction network (VAIN) [48] introduced an attention mechanism for predictive modeling of multiagent interactions, in which each agent can selectively receive information from other agents. Sukhbaatar et al. [43] designed CommNet, a simple neural model that differs from earlier approaches [49], [50] of manually predefined communication protocols by allowing the network to autonomously learn how to communicate with each agent, enabling simpler and more effective collaboration. Zou et al. [47] proposed import message attention network (IMANet) to address the challenge of informational coordination, which divides the process of making decision into two phases: communication and action. Recently, some researchers have focused more on collaborative perception [51], [52], [53] tasks in multiagent communication systems. For example, in autonomous driving systems, Wang et al. [54] used vehicle-to-vehicle (V2V) communication to aggregate information received by multiple vehicles surrounding them as a way to enhance the perception and prediction capabilities of self-driving vehicles. Xu et al. [55] expand the targets of collaboration beyond vehicles to infrastructures, combining the spatial position relationships of each agent to determine the agent interaction process.

Observation platforms in RS can also be regarded as independent intelligent agents, and several deep neural networks are already able to be deployed on RS platforms for Earth observation tasks. However, according to our research, there has been no further exploration related to collaboration for multiple RS platforms. Actually, the same limitations of a single-platform's observation exist under some observation conditions as well. In order to fill this gap, it is necessary for us to investigate the approaches to collaborative inference for multiple RS platforms.

III. Proposed Method

A. Overview

The overall framework of the method is illustrated in Fig. 2. The framework process can be separated into four stages: feature extraction, collaborative request generation (request), collaborative request matching (match), features transmission, and information fusion (fusion). The platform with a task execution requirement is referred to as the requesting end, while other platforms are referred to as the candidate ends. Each end employs YOLOv5l as the basis object detection model, and it can be used as a typical independent inference, ensuring that the platform can still execute the task in the absence of collaboration response to the request.

The MDCNet implements the collaborative inference process as follows.

1) Feature Extraction: First, we use the backbone of a basis detection network to perform feature extraction on the input image. This step is performed locally on each platform.

2) Request and Match: Specifically, the requesting end can match its own requirements with the current information of other ends and then connect with the end with the highest degree of matching. In this step, the requesting end generates a spatial collaboration map for spatially sparse collaborative request map encoding and broadcasts it to others. Next, the candidate ends that receive the collaborative request map will

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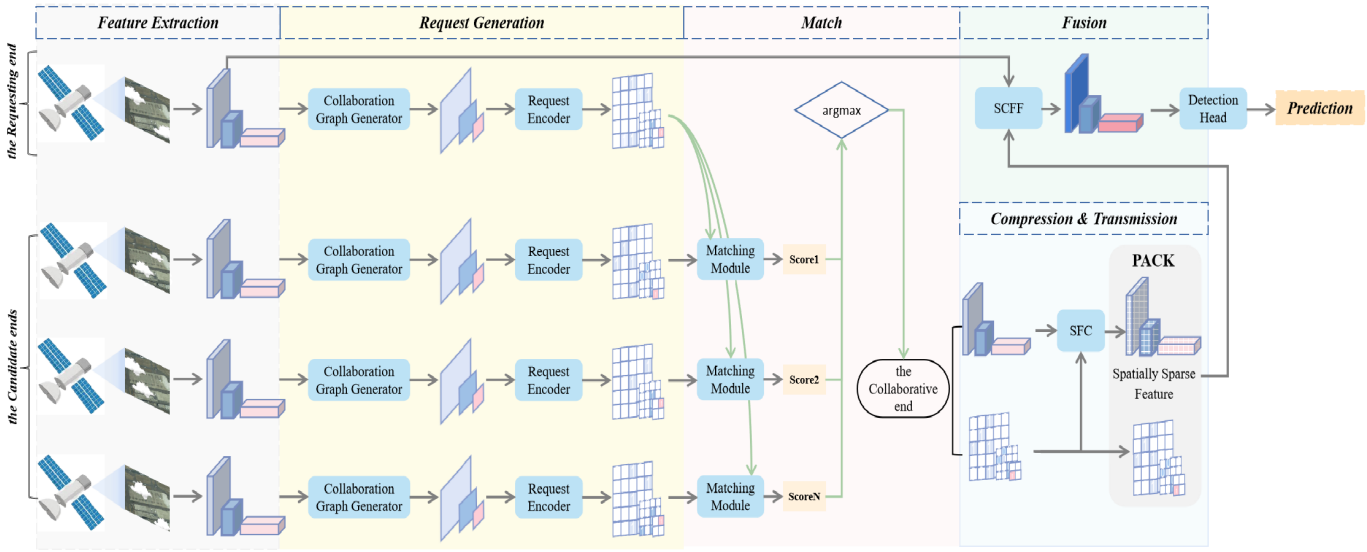


Fig. 2. Overview of the proposed framework. The whole collaborative inference process can be divided into four parts: 1) feature extraction: each end extracts the features of the input image; 2) request: the requesting end generates and broadcasts a collaboration request; 3) match: the candidate ends calculate the matching scores and send them back; then the requesting end selects the collaborative end with the highest score; and 4) transmission and fusion: the collaborative end sends the feature compressed by SFC module to the requesting end. Then, the proposed SCFF module guides the feature fusion process at the requesting end.

calculate the matching scores locally and send the scores back to the requesting end. Then, the requesting end will select the one with the highest matching score of the multiple candidate ends to establish a collaborative connection.

3) Transmission and Fusion: In the collaborative inference, information is exchanged between different ends through the interaction intermediate feature maps. Meanwhile, as we know, the majority of the spatial areas mapped by the intermediate feature maps may be object-free background areas in the RS object detection task. However, during multiplatform collaboration, this redundant information will consume significant communication costs, but generate little information gain. Instead, we should pay more attention to those areas containing objects that can help supplement the object information not detected by other platforms in the inference, which can generate information gain for the collaborative inference. As a result, an effective method called the SFC module aims to compress the intermediate feature maps before transmission. The requesting end receives the message from the collaborator, which consists of the collaborator's compressed features and the collaborative request map. Afterward, the collaborative request map is decoded into the original spatial collaboration map at local. Finally, the SCFF module utilizes the spatial collaboration map as the spatial priors for the subsequent multiscale feature fusion to make a better fusion.

After that, the updated fused features are input to the local detection head, which outputs the final detection results.

B. Baseline Architecture

Considering the trade-off between inference speed and accuracy, we employ the YOLOv5l network as the baseline model in our framework, which refers to [56] for the deployment of advanced deep neural network real-time object detection algorithms on edge and mobile devices. In general, it is composed of two primary parts: the backbone and the head (containing the neck). The backbone aims at extracting low-level geometric features and high-level semantic features, which use the path aggregation network (PAN) [57] and FPN [58] to recombine the feature maps extracted in the three stages. This structure builds an improved FPN by incorporating features from top to bottom, supplies robust semantic features, and propagates the underlying texture and pattern features from bottom to top. At last, the new multiscale features obtained after the recombination are input into the head to generate detection results.

C. Collaboration Request Generator

In order to select the optimal collaborator and maximize the information gain, we propose a collaboration map generator module that employs a detection decoder responsible for decoding the intermediate layer features and generate a spatial collaboration map of the current scale feature map. The generated map reflects the confidence level of whether each position of the feature map contains an object. For the ith platform, the feature maps are represented by $f_i^l \in \mathbb{R}^{C^l \times H^l \times W^l}$ at the lth scale, $l = 1$–3, corresponding to the three stages of YOLOV5l's feature extraction, C^l represents the channel number, and H^l and W^l denote the height and width of feature map at the lth scale, respectively. The corresponding spatial collaboration map s_i^l is generated as follows:

$$s_i^l = G\left(f_i^l\right) \tag{1}$$

where $G(\cdot)$ denotes the spatial collaboration map generator, which is essentially a detection decoder. The value of the elements in $s_i^l \in \mathbb{R}^{H^l \times W^l}$ ranges from 0 to 1, indicating the confidence level of the object existence at each position in the current scale feature map. In particular, a higher score for corresponding position of the spatial collaboration map indicates that there is a higher possibility of the existence of the object in the mapped area, whereas a lower score means that it cannot determine whether the area contains an object. For the ith platform, the spatial collaboration map reflects which areas are available for sharing and performing the collaborative inference with other platforms, and which

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areas may have missing information and need to ask for help from others.

The spatial collaboration map generated by the above generator based on the feature maps reflects the spatial confidence level for the input image at a specific feature scale. For spatial regions with low confidence levels, there is greater uncertainty about the presence or absence of objects. Also, this uncertainty may be caused by the presence of an abnormally occlusion, which is an area of information lacks for the current platform. At this time, the current platform sends a collaboration request to seek the other platforms' information for the area of low confidence, after which it reintegrates the information and updates the detection result. As a result, the collaboration request map is defined as follows:

$$r_i^l = 1 - s_i^l. \tag{2}$$

During the collaborative inference process, the requesting end first broadcasts its collaboration request to the neighboring ends. Then, the end receiving the request will calculate the matching score between the collaboration request from the requesting end and the spatial collaboration map generated by itself, which maps the degree of complementarity of the difference information between the two work ends. Subsequently, after receiving the score from the other end, the requesting end will establish a connection with the end with the highest matching score and prepare for feature interaction. In order to complete the collaboration request matching more directly, the requesting end encodes the collaboration request map into a binary matrix when broadcasting the request. Define the request message sent from the ith requesting end to the other end as M_i, and the matching score G_{ji} sent back from the jth end is calculated as follows:

$$m_i^l = 1 - c_i^l \tag{3}$$

$$c_i^l = \phi_{\text{BE}}\left(s_i^l\right) \tag{4}$$

$$g_{ji}^l = \Phi_{\text{score}}\left(c_j^l \circ m_i^l\right) \tag{5}$$

$$G_{ji} = \sum_l g_{ji}^l \tag{6}$$

where $\phi_{\text{BE}}(\cdot)$ denotes the binary conversion function, and in the generated binary selection matrix $\phi_{\text{BE}}(s_i^l)$, element 1 indicates that the object detection confidence is greater than or equal to the set threshold, and the corresponding spatial area mapped by the position contains the object to be detected. Conversely, element 0 indicates that the object detection confidence is lower than the set threshold, representing the lack of regional information.

Conversely, in m_i^l, element 1 indicates the ith end's request for supplementary information at that area, while element 0 indicates confident detection results for the corresponding area. In the matching score calculation, $\circ$ is the elementwise product of the matrix, and $\Phi_{\text{score}}(\cdot)$ measures the degree of matching between the binary matrix m_i^l of collaborative requests sent from ith end to jth end and the corresponding spatial collaboration map of the jth end. Specifically, at each position in $c_j^l \circ m_i^l$, element 1 indicates a successful request matching, and element 0 indicates a failed request matching. Summing the elements of all positions measures the degree of matching of the entire map. Therefore, a higher synthesized score indicates a higher degree of matching. Finally, the end with the highest matching score is selected as the collaborative end

$$j_c = \arg\max_j G_{ji}. \tag{7}$$

D. SFC Module

RS images, in contrast to natural images, have a broad field of view and complex objects of diverse scales. The YOLOv5l object detection model used in this article, whose backbone has three scale feature output branches, can detect objects at different scales in RS images. Similarly, in the process of multiplatform collaborative inference, we will simultaneously fuse the feature-level data of three different scales to realize the information complement of multiscale objects, thereby improving the overall detection accuracy. However, considering that multiscale feature transmission will exponentially increase the transmission cost, we present a spatially sparse feature compression method. The SFC module lowers the transmission cost between collaborative platforms by compressing a large number of mapped object-free background areas before transmitting the intermediate feature maps. Specifically, after establishing the collaborative connection, the collaborative end j_c selects the areas with the highest object confidence in the local feature map $f_{j_c}^l$ as the information to be transmitted under the guidance of the binary selection matrix $c_{j_c}^l$ generated by itself. Specifically, the feature map to be transmitted to ith end at the lth scale is selected as follows:

$$F_{j_c,i}^l = c_{j_c}^l \circ f_{j_c}^l. \tag{8}$$

Since many positions of $c_{j_c}^l$ have element 0, the generated $F_{j_c,i}^l$ also has element 0 at the corresponding position, which is sparser spatially and facilitates compression during subsequent transmission.

In addition to the compressed feature map, the collaborative end j_c also transmits its own collaboration request M_{j_c} to obtain the collaborative information and to provide spatial priors for the later feature fusion process occurring at the requesting end. In a nutshell, the information to be sent from the collaborative end to the requesting end is

$$N_{j_c,i} = \left(M_{j_c}, F_{j_c,i}\right) \tag{9}$$

$$M_{j_c} = \text{Stack}\left[m_{j_c}^1, m_{j_c}^2, m_{j_c}^3\right] \tag{10}$$

$$F_{j_c,i} = \text{Stack}\left[F_{j_c,i}^1, F_{j_c,i}^2, F_{j_c,i}^3\right]. \tag{11}$$

Here, due to the introduction of the binary selection matrix, many portions of the multiscale intermediate feature maps are zeroed and transmitted, resulting in very sparse feature compression and a significant reduction in transmission cost during the collaborative inference.

E. SCFF Module

After receiving the message N_{j_c}, the requesting end first decodes the collaboration request M_{j_c} into its corresponding binarized spatial collaboration map C_{j_c}. Specifically, $C_{j_c} = 1 - M_{j_c}$, it is a matrix elementwise operation for all scales. For the received spatially sparse feature maps, those zero-setting areas will negatively interfere with the fusion process if we fuse them directly. Therefore, we employ the spatial

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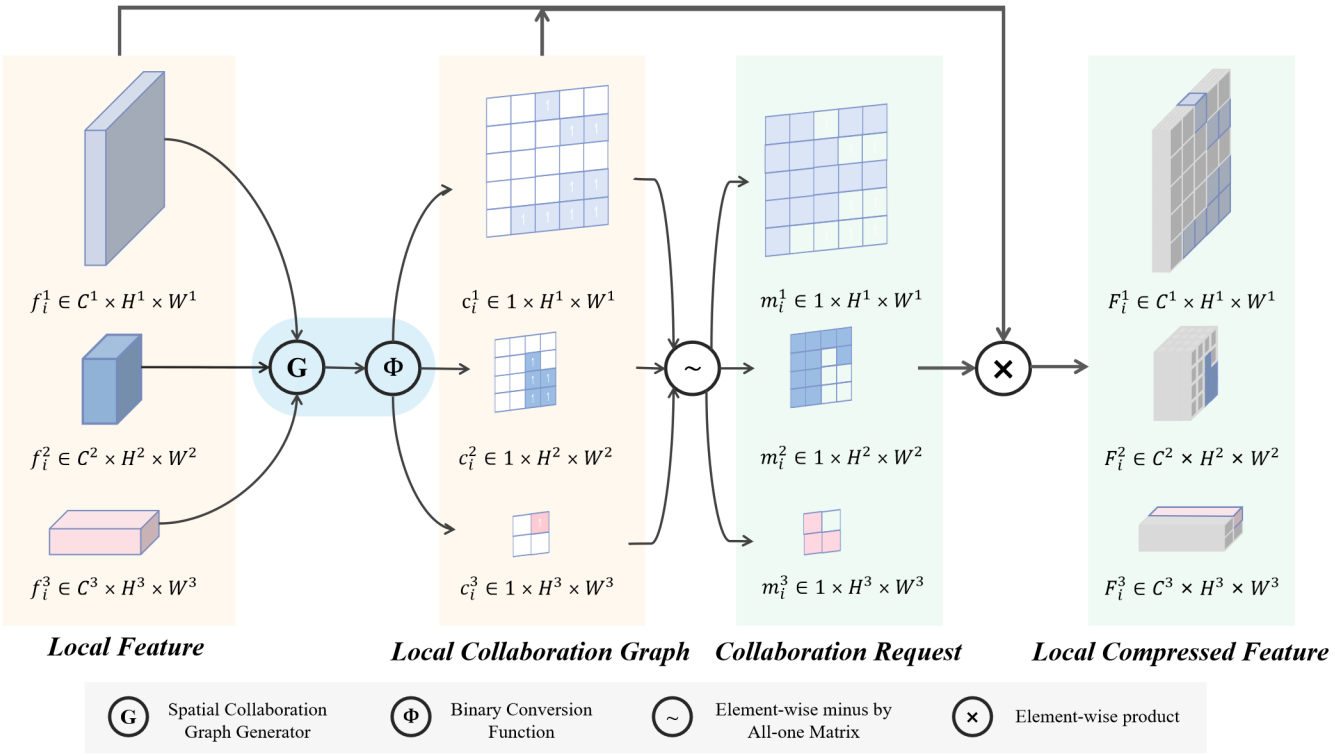


Fig. 3. Schematic of the SFC module. The backbone of YOLOv5l for feature extraction produces different feature maps at three scales, and the computational process in the figure includes the generation of the local spatial collaboration map, the collaboration request map, and the compressed spatially sparse features.

collaboration map C_{j_c} as spatial priors in the multiscale feature fusion process and perform location-by-location feature fusion on the feature maps. Then, we use a multihead attention for each scale. In detail, it simultaneously inputs the feature maps f_i^l of the requesting end, the compressed feature maps $F_{j_c,i}^l$, and the spatial priors $c_{j_c}^l$ provided by the collaborative end and performs the location-by-location fusion on the feature maps of the three scales in parallel. After the above operation, the feature maps of the ith end at the lth scale are updated as follows:

$$f_{i_{\text{fusion}}}^l = \text{mha}^l\left(f_i^l, F_{j_c,i}^l, c_{j_c}^l\right) \tag{12}$$

where $\text{mha}^l(\cdot)$ is the position-by-position multihead attention at the lth scale and $c_{j_c}^l$ is used as a mask in the multihead attention fusion process to shield the spatially zero area from the fusion of other areas. Finally, the updated feature maps are input to the detection head, and the prediction results are as follows:

$$\hat{y} = D_i(f_{i_{\text{fusion}}}; \theta_d) \tag{13}$$

where $D_i(\cdot)$ indicates the detection head of the ith end, $f_{i_{\text{fusion}}} = \text{Stack}[f_{i_{\text{fusion}}}^1, f_{i_{\text{fusion}}}^2, f_{i_{\text{fusion}}}^3]$, and θ_d represents the parameters of the detection head.

IV. Experiments and Analysis

We evaluate the performance of our designed multiplatform collaborative network on three public datasets in this section. In specific, this section is organized into the following three parts: first, we introduce the datasets and evaluation metrics used in the following experiments. Then, we assess the performance of the components of the method through ablation experiments. Finally, the superiority of the proposed framework is verified intuitively through results visualization.

A. Datasets and Experimental Settings

Since the current publicly available RS object detection datasets [59], [60], [61], [62], [63] only reflect single-platform observations, there is a lack of corresponding datasets for multiplatform collaborative inference. Therefore, the experimental datasets in this article are constructed using three different RS datasets, called NWPU VHR-10 [59], DOTA [60], and FAIR1M [61], to synthesize multiplatform collaborative RS datasets as experimental support. Then, we use the collaboration of four RS platforms to verify the effectiveness of the proposed framework.

1) Basic Datasets:

a) NWPU VHR-10: NWPU VHR-10 is published by Cheng et al. [59] in 2014, which was public and available for horizontal RS object detection from Google Earth and Vaihingen. The dataset comprises 800 very high-resolution (VHR) RS images in total, of which 80% contain ten types of objects, as shown in Table I, for a total of 3651 objects, and the remaining 150 are background images that do not contain any objects. We only use the 650 images containing objects, of which 80% are randomly designated for training purpose and the remaining are reserved for testing.

b) DOTA: DOTA is enormous and primarily utilized for RS object detection, containing 2806 images from various sensors and platforms, each with pixel size ranging from 800 × 800 to 4000 × 4000, involving 15 categories. The categories defined in this dataset are shown in Table I. In the following experiments, the original images are cropped to obtain a uniform image size of 1024 × 1024. Then,

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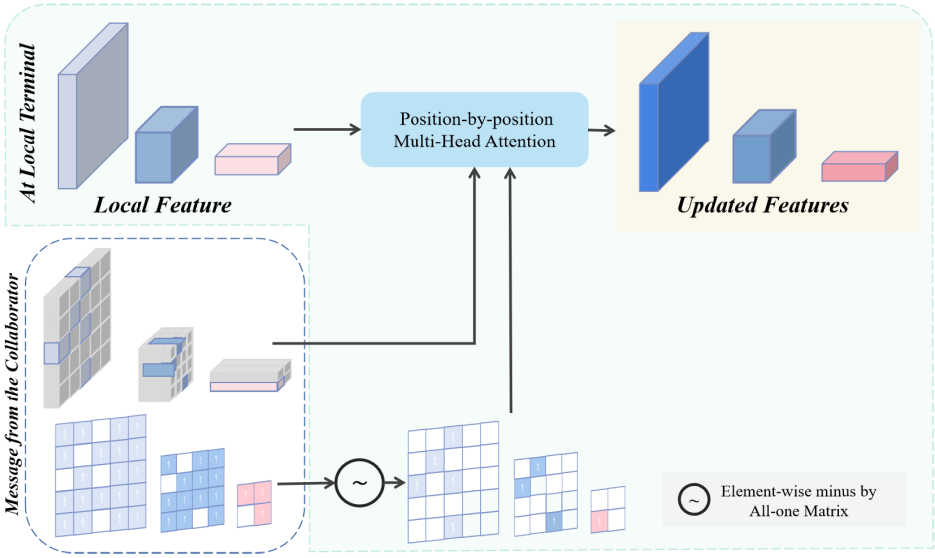


Fig. 4. Schematic of the SFFC module. After receiving the message back from the collaborative end, feature fusion guided by spatial priors is performed at the local platform to update the features.

TABLE I
CATEGORY ABBREVIATION

NWPU VHR-10		DOTA	
Full Name	Abbreviation	Full Name	Abbreviation
Airplane	APL	Small Vehicle	SV
Ship	SH	Large Vehicle	LV
Storage Tank	ST	Plane	PL
Baseball Diamond	BD	Storage Tank	ST
Tennis court	TC	Ship	SH
Basketball Court	BC	Harbor	HA
Ground Track Field	GTF	Ground Track Field	GTF
Harbor	HA	Soccer Ball Field	SBF
Bridge	BR	Tennis court	TC
Vehicle	VE	Swimming Pool	SP
		Baseball Diamond	BD
		Roundabout	RD
		Basketball Court	BC
		Bridge	BR
		Helicopter	HE

2500 randomly cropped images are selected for training, while 800 are allocated for the test set.

c) FAIR1M: FAIR1M, proposed in 2021, is recently the world's largest public dataset for fine-grained object detection in RS. The resolutions of each image is as high as 0.3–0.8 m. The scenes in this dataset are distributed in several countries and regions, involving five object categories: aircraft, ships, vehicles, sites, and roads, with a total of 37 fine-grained categories, including 11 fine-grained categories for aircraft, nine specific ship-specific categories, ten vehicle categories, four site categories, and three road categories. With multiple object scales, arbitrary orientations, and varied fine-grained categories, this dataset is a challenging detection dataset. We select some of these image data in this dataset for experimental validation.



Fig. 5. RS images from different platforms in the case of thin cloud blurring.

2) Experimental Settings: When data from a single platform degrade, it is necessary to obtain complete observation information by merging information from multiple platforms. In Earth observation by RS platforms, cloud interference is a common cause of observation data degradation [64], [65]. Therefore, our experimental setting is designed to synthesize data degradation with two different degrees of cloud occlusion, so as to verify the effectiveness of the proposed multiplatform framework.

a) Thin Cloud Blurring: Referring to [66], we use the summation of multiscale random noise images to generate synthetic thin-cloud blurred images I_{thin}. The specific synthesis is calculated as follows:

$$I_{\text{thin}} = \sum_{s} \Psi(\text{Rand}(2^s))/2^s \tag{14}$$

where $\text{Rand}(2^s)$ denotes a random function generating an image with random noise of size 2×2 and $\Psi(\cdot)$ denotes an operator adjusting the random noise to the size of the original image. Then, s denotes a scale factor whose value is a natural number from 1 to $\log_2(N)$, and N is the size of the original image. In the experimental setting of this article, and s is set to 2. After generating and accumulating two random noises of equal image size consecutively, the noise interference is overlaid on the original images to generate the new images. An example of a set of images in a synthetic dataset is shown in Fig. 5.

b) Thick Cloud Occlusion: By overlaying opaque cloud masks onto different regions of the image, this process simulates the impact of dense atmospheric clouds on satellite platforms during the acquisition of RS images. All opaque cloud masks are derived from real cloud images. When synthesizing images from different platforms of observation, each

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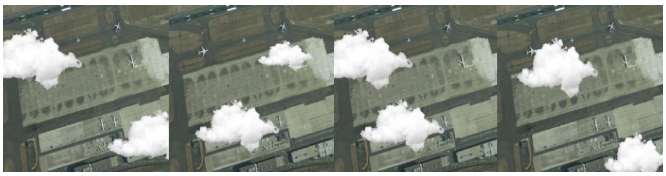


Fig. 6. RS images from different platforms in the case of thick cloud occlusion.

image contains two cloud masks. The areas obscured by the cloud masks between the different platforms do not overlap at all or partially overlap to ensure that there is a difference in the degree and position of missing information in the images of the different platforms. An example of a set of resulting synthesized images is depicted in Fig. 6.

The two settings described above depict distinct scenarios encountered in Earth observation. In the first scenario, the presence of thin clouds introduces a degree of blurring to the object features, which can impact the accuracy of object detection. However, even under cloud cover, the weak object features are still partially retained, and they are not entirely lost. Consequently, there is a possibility of obtaining some information regarding the features, albeit with reduced clarity. In contrast, in the case of thick clouds, the features beneath the cloud cover are completely obscured, rendering the information effectively zero. In such case, if we solely rely on single-platform inference, the obtained information at the cloud position will inevitably be zero as well. In contrast, despite the fact that information at each end of the above two setups is lacking, the distribution of thin clouds and the extent of thick cloud cover observed at different ends are different due to the variations in observation angles and moments. As a result, the information between the different ends can be overlapping or complementary, and the fusion of the complementary information can lead to more comprehensive information about the target area.

B. Evaluation Metrics

In our experiments, we evaluate the network's performance using two metrics: accuracy metrics and transmitted data quantities.

1) Accuracy Metrics: The average precision (AP) is a widely utilized assessment measure for quantitatively analyzing the object detector's capabilities by assessing the differences in agreement between the prediction results and the ground truth. In general, the output of the detector is classified into four categories: true positive (TP), false negative (FN), true negative (TN), and false positive (FP), where TN and TP represent precise predictions and FN and FP indicate imprecise predictions. Counting the number of samples from each of the four cases independently, the precision and recall may be calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{16}$$

As the above equation shows, precision and recall are contradictory to each other. Therefore, the area of the precision–recall (PR) curve in each recall case from 0 to 1 is defined as the AP to balance the two metrics. For multiclass object detection, mean AP (mAP) is a composite metric computed by averaging APs across all classes and can be computed as follows:

$$\text{mAP} = \frac{\sum_{i=1}^{C} \text{AP}_i}{C} \tag{17}$$

where AP_i denotes the AP of ith class and C denotes the total number of classes.

2) Transmitted Data Quantities: Due to the constraints of interstellar transmission conditions in multiplatform collaboration, it is important to control the amount of data transmitted between platforms as small as possible. The most direct approach is to transmit the original images for centralized processing. Consequently, we take the size of the original image, denoted as $W \times H$, as a benchmark. The transmission cost ratio in subsequent experiments can be computed using the following equation:

$$\text{comm_ratio} = \frac{\sum_{l=1}^{L} \sum_{i=1}^{w_l} \sum_{j=1}^{h_l} c_{ij}^{l}}{W \times H} \tag{18}$$

where W and H represent the width and height of the original input images, respectively, while w_l and h_l denote the size of the intermediate feature map at the lth scale. Also, c^l denotes the binary selection matrix at the lth scale in our method, which takes the values 0 and 1. The above equation calculates the amount of the nonzero section of the final transmitted intermediate feature maps, which accounts for a major portion of the transmission cost.

C. Implementation Details

The framework proposed in this article is implemented in PyTorch 1.9 and executed on an RTX-2080Ti GPU. To assess the effectiveness of the above framework, during following experiments, we employ the same detection model structure, YOLOV5l, at each platform. First, we pretrain the detection model on the corresponding dataset and then apply this model to our proposed collaborative framework. In specific, during the training stage, the detection model's parameters at each platform are frozen and are not involved in later updates. This is to ensure that the gains are all derived from the collaborative framework, independent of the detection capability of the network at each platform. In addition, in the training and testing stages, the input image's size is standardized to 640 × 640. Also, during the training stage, with 250 epochs of training rounds, the standard stochastic gradient descent (SGD) [67] optimizer is employed, with the momentum configured at 0.937 and the weight decay at 0.0005. Meanwhile, the learning rate is set to 0.001 at the beginning.

D. Ablation Study

First, we design a series of ablation experiments on the validation set and validate the proposed method's efficacy under two different experimental scenarios based on the NWPU VHR-10 dataset and the DOTA dataset.

1) Validation of the Baseline Multiplatform Collaborative Inference Framework: In Tables II and III, the accuracy upper bound for all methods is the result of normal and clear images input to the original network for forward inference. For the NWPU dataset and the DOTA dataset, the upper

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TABLE II

COMPARISON OF INFERENCE ACCURACY BETWEEN SINGLE-PLATFORM AND MULTIPLATFORM COLLABORATIVE NETWORK BASED ON NWPU VHR-10. (A) NWPU VHR-10-THICK. (B) NWPU VHR-10-THIN

(a)

Source of Inference (SOI)		APL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mAP_{50}	AV	Δ
Clear Image[UpperBound]		99.5	92.9	98.7	98.3	90.7	84.1	99.5	96.4	35.5	74.9	87.1	87.1	-
Single platform 1		82.7	67.5	88.3	83.5	67.4	71.0	71.7	58.9	30.8	57.9	68.0	68.90	-
Single platform 2		77.3	76.4	69.9	78.8	79.0	73.0	91.4	68.8	32.9	50.8	69.8		
MDCNet	platform 1	**98.8**	**87.0**	**95.7**	**96.7**	**87.1**	**82.4**	**94.7**	**62.6**	**33.8**	**78.3**	**81.7**	**82.85**	↑ 13.95
	platform 2	**99.2**	**89.2**	**94.9**	**96.3**	**88.6**	**82.2**	**99.5**	**82.0**	**32.3**	**76.3**	**84.0**		

(b)

Single platform 1		59.1	59.5	93.6	77.8	7.94	3.94	61.7	41.4	26.9	18.2	45.0	46.75	-
Single platform 2		54.7	64.3	94.1	78.6	17.5	25.7	70.6	39.5	12.4	27.3	48.5		
MDCNet	platform 1	**97.0**	**84.2**	**94.2**	**92.8**	**64.6**	**71.8**	**90.3**	**73.3**	**25.5**	**59.4**	**75.3**	**75.60**	↑ 28.85
	platform 2	**96.0**	**81.5**	**91.5**	**92.6**	**64.7**	**71.0**	**89.8**	**73.4**	**38.0**	**60.8**	**75.9**		

TABLE III

COMPARISON OF INFERENCE ACCURACY BETWEEN SINGLE-PLATFORM AND MULTIPLATFORM COLLABORATIVE NETWORK BASED ON DOTA. (A) DOTA-THICK. (B) DOTA-THIN

(a)

Source of Inference (SOI)		SV	LV	PL	ST	SH	HA	GTF	SBF	TC	SP	BD	RD	BC	BR	HE	mAP_{50}	AV	Δ
Clear Image[UpperBound]		65.5	86.5	92.7	69.2	88.5	83.0	56.6	59.8	92.9	68.8	80.6	59.3	51.7	48.5	35.1	69.2	69.2	-
Single platform 1		55.4	75.9	77.3	29.6	77.8	64.1	38.1	47.7	75.1	56.5	64.7	34.1	37.9	39.5	27.7	53.4	54.6	-
Single platform 2		57.8	73.4	82.8	43.6	77.9	63.2	45.0	47.9	77.4	56.0	61.6	41.9	50.6	34.8	22.8	55.8		
MDCNet	platform 1	**56.2**	**79.0**	**85.7**	**59.4**	**84.1**	**72.6**	**34.4**	**49.6**	**89.3**	**60.4**	**63.5**	**35.7**	**51.2**	**38.2**	**19.2**	**58.6**	**59.15**	↑ 4.55
	platform 2	**58.0**	**78.1**	**86.7**	**60.5**	**84.8**	**73.0**	**40.7**	**53.8**	**90.6**	**57.5**	**62.3**	**38.5**	**55.7**	**36.5**	**18.9**	**59.7**		

(b)

Single platform 1		44.1	66.0	82.0	54.5	82.4	63.9	14.8	32.0	83.9	0.0	28.8	26.2	30.7	21.9	10.2	42.8	43.05	-
Single platform 2		43.9	63.8	81.9	56.4	82.0	62.4	17.7	39.1	85.3	0.0	31.4	23.5	36.1	19.9	5.7	43.3		
MDCNet	platform 1	**56.0**	**78.2**	**85.7**	**59.8**	**84.1**	**74.2**	**31.5**	**46.3**	**87.7**	**33.8**	**44.7**	**39.8**	**40.1**	**30.8**	**36.8**	**55.3**	**55.05**	↑ 12.00
	platform 2	**55.1**	**77.9**	**85.2**	**60.7**	**83.8**	**73.6**	**29.3**	**44.3**	**89.1**	**32.5**	**51.5**	**35.9**	**39.0**	**29.4**	**34.1**	**54.8**		

bounds are 87.1% and 69.2%, respectively. Furthermore, the table lists the inference performance of the single-platform independent network and the proposed collaborative inference framework, as assessed by the evaluation metric mAP_{50}, whose intersection over union (IoU) threshold is set to 0.5.

a) Results in the case of thick cloud occlusion: As the results on the NWPU dataset shown in Tables II(a) and III(a), the average accuracy of single-platform inference decreases to 68.90%, representing a decrease of 18.20% compared with the upper bound, which is taken as the lower bound. After applying the proposed multiplatform distributed collaborative inference framework, the detection accuracy is improved at each platform. Specifically, the average result is 82.85%, 13.95% higher than the lower bound. Similarly, on the DOTA dataset, the lower bound is 54.6%, decreasing by 14.60%. When employing the proposed collaborative framework, the result achieves an improvement of 4.55%. Also, as reported in the last row of table, compared with single-platform inference, the proposed method consistently achieves improvement in almost all classes.

b) Results in the case of thin cloud blurring: As listed in Tables II(b) and III(b), under thin cloud interference conditions, the lower bounds based on the NWPU dataset and DOTA dataset are 46.75% and 43.05%, respectively. Compared with single-platform inference, the proposed method improves mAP on the NWPU dataset by 4.55%, while it improves mAP by 12.00% on the DOTA dataset as well.

In both cases mentioned above, when single-platform object detection interfered and results degraded, the proposed framework clearly yields accuracy gains for both collaborative platforms. This verifies the effectiveness of the proposed collaborative approach and demonstrates that information sharing between the collaborative platforms during the collaborative inference process achieves an effective supplement to the region of insufficient local information.

2) Modules Analysis Within the Multiplatform Collaborative Inference Framework: As described in Section III, the proposed framework contains three key components: collaborative request matching module, SFC module, and SCFF module. We conduct a series of ablation experiments on the NWPU validation set in the case of thick cloud occlusion to analyze the contribution of each module.

a) Effect of collaborative request matching module: The first step in multiplatform collaborative inference is the selection of the collaborative platform that can achieve the maximum information gain in the subsequent collaboration.

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TABLE IV
ABLATION STUDIES FOR COLLABORATIVE REQUEST MATCHING MODULE

MODEL	APL	SH	ST	BD	TC	BC	GTF	HA	BR	VE	mAP_{50}	Δ
Single platform [LowerBound]	82.7	67.5	88.3	83.5	67.4	71.0	71.7	58.9	30.8	57.9	68.0	–
Baseline$^-$	95.1	80.3	93.2	92.3	85.8	73.3	92.6	60.6	33.5	75.9	78.1	↑10.1
Baseline$^+$	**98.8**	**87.0**	**95.7**	**96.7**	**87.1**	**82.4**	**94.7**	**62.6**	**33.8**	**78.3**	**81.7**	**↑13.7**

TABLE V
ABLATION STUDIES FOR SFC MODULE

MODEL	mAP_{50}	Δ	Comm-ratio
LowerBound	68.0	-	0
SFC$^-$	82.1	↑ 14.1	1/1
SFC$^+$	81.7	↑ 13.7	0.1624/1

TABLE VI
COMPARISON WITH OTHER FUSION METHODS

Method	NWPU-thick		NWPU-thin	
	mAP_{50}	Δ	mAP_{50}	Δ
LowerBound	68.0	-	45.0	-
+ATTEN	71.0	↑ 3.0	46.9	↑ 1.9
+MAX	73.4	↑ 5.4	47.9	↑ 2.9
ours	**81.7**	**↑13.7**	**75.3**	**↑10.3**

Therefore, we propose a collaborative request matching module, which selects the best collaborator among the candidate ends by matching the collaboration request maps between the requesting end and other ends. The ablation results of this module are shown in Table IV, comparing the results of random selection (randomly select the collaborator) and applying the proposed matching module, denoted by Baseline$^-$ as well as Baseline$^+$, respectively. As the results show, random selection yields an accuracy gain compared with the lower bound of single-platform inference, representing a 10.1% improvement in the mAP_{50} metric. This improvement demonstrates the contribution of the collaborative framework as well, and there exists a 1/3 probability of selecting the best match during the random selection. Besides, after the addition of the collaborative request matching module, which selects the best collaborator for each collaboration, the detection accuracy of all classes compared with random selection is improved, and the mAP_{50} increases by 3.6%.

b) Effect of SFC module: In the case of poor intersatellite communication, we design the SFC module to compress the information transmission during collaboration, transmit less collaborative information, and obtain the maximum inference gain. For the object detection of RS images, the part of the feature map that characterizes the background information has a high transmission occupation but is less useful for object detection. The SFC module proposed in this article achieves the reduction of information transmission by compressing this part, and SFC$^+$ and SFC$^-$ in Table V indicate whether the compression module is used or not, respectively.

According to the results, it can be shown that without applying the SFC module, the mAP_{50} is improved by 14.1% after multiplatform collaboration compared with the lower bound, but the communication cost ratio is 100%, while when using the SFC module, the mAP_{50} is only lost by 0.4% compared with the full transmission, and the communication cost ratio is 16.24% of the full transmission. This verifies that the module trades a small loss of accuracy for more than 80% of the transmitted information compression.

Fig. 7 shows the comparison of the saliency maps of the intermediate features before and after fusion using the proposed method, which visually illustrates the effect of collaborative fusion on multiple platforms. We visualized the saliency maps generated by the YOLO network. Also, the different color depths in the saliency maps reflect the learning focus of the network on each intermediate feature. The figure shows that the network is able to warn about objects under the clouds (outlined in a red circle) with a more focused focus after using the method proposed in this article.

3) Comparative Experiments With Other Feature Fusion Methods: Because the transmitted intermediate features are spatially sparse after the SFC module, the spatial prior information reflected by the sparse feature map should also be considered in the feature fusion stage. Therefore, we propose the SCFF module in this article and consider two common basic feature fusion approaches as the baseline model for comparison, namely, attention and MAX.

In Table VI, we list the detection results obtained by various fusion modules in multiplatform collaborative framework in both case of experimental settings. The results show that using the fusion module in this framework improves the detection accuracy compared to single-platform inference, and compared with the basic fusion modules attention and MAX, the proposed SCFF module further improves the detection accuracy by 10.7% and 8.3% in the case of thick cloud occlusion and by 8.4% and 7.4% in the case of thin cloud blurring, respectively.

E. Effect of Transmission Threshold Variation on Detection Accuracy and Transmission Cost

The hyperparameter transmission threshold affects the spatial sparsity of the transmitted feature maps, and the portion of the original feature maps with greater confidence scores than the threshold is provided for the collaborator. Therefore, we further analyze the effect of this hyperparameter on the detection accuracy and the transmission cost between the collaborators in the two experimental settings. As shown in Table VII, by adjusting the transmission threshold from 0.01 to 1.0, it is observed that the cost of transmission decreases exponentially as the transmission threshold increases, but the

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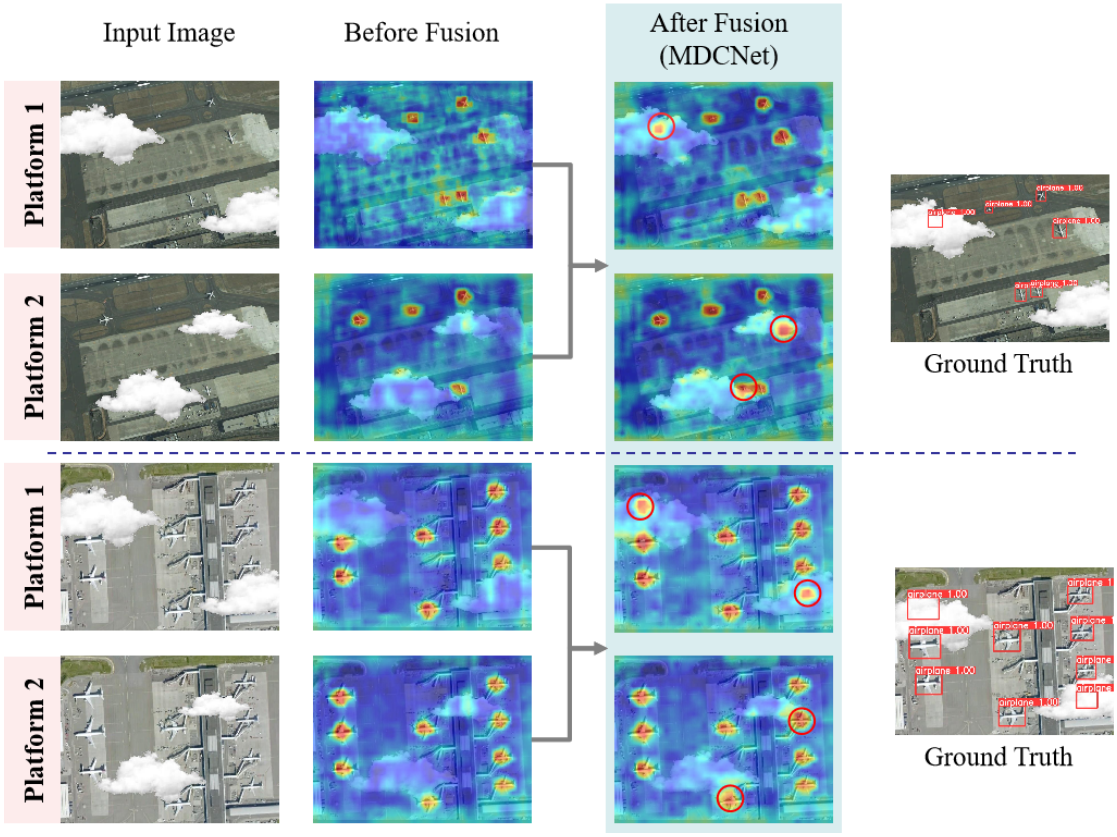


Fig. 7. Comparison of visualized saliency maps of intermediate features before and after the proposed multiplatform collaborative fusion.

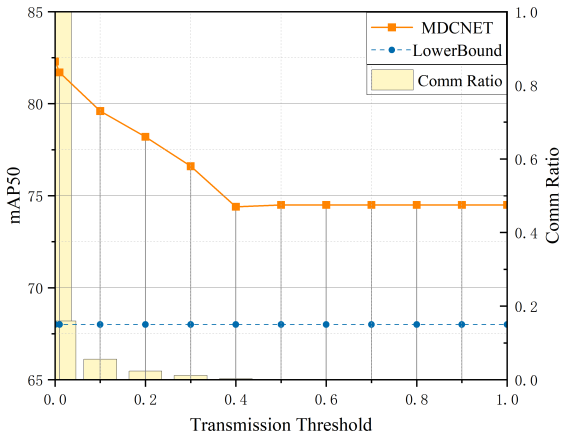


Fig. 8. Effect of transmission threshold variation on detection accuracy and transmission cost in the case of thick cloud occlusion.

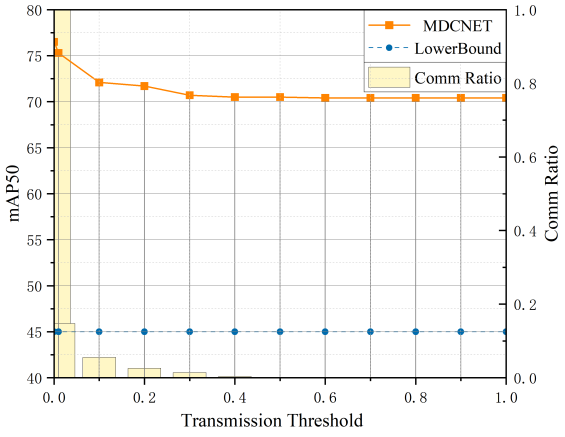


Fig. 9. Effect of transmission threshold variation on detection accuracy and transmission cost in the case of thin cloud blurring.

accuracy also decreases. Also, as represented in Figs. 8 and 9, we find that when the threshold is greater than 0.4 and then continues to increase, the transmission between collaborators is almost zero at the same time. However, the accuracy of our method is still higher than the lower bound, which is due to the fact that the multiheaded attention mechanism used in the feature fusion module of this framework itself also enhances the intermediate features to achieve inference gain.

F. Results on the Difficult Dataset Fair1M

To further validate the capability of the proposed framework, we also conducted experimental validation in the thin cloud case on the FAIR1M dataset, which has a richer number of categories and a larger number of objects and contains 37 fine-grained categories with a more challenging task. As the experimental results show in Table VIII, under interference, the detection accuracy of single-platform inference drops to 24.95%. But, after applying the collaborative method proposed in this article, the average accuracy of detection is improved by 3.1%, which validates the superiority of the proposed framework.

G. Visual Analysis

In this section, we select several different scenarios to compare the visual detection results of single-platform independent inference and the multiplatform collaborative inference introduced in this article, as depicted in Fig. 10. It can be observed that the single platform does not have the ability to detect the

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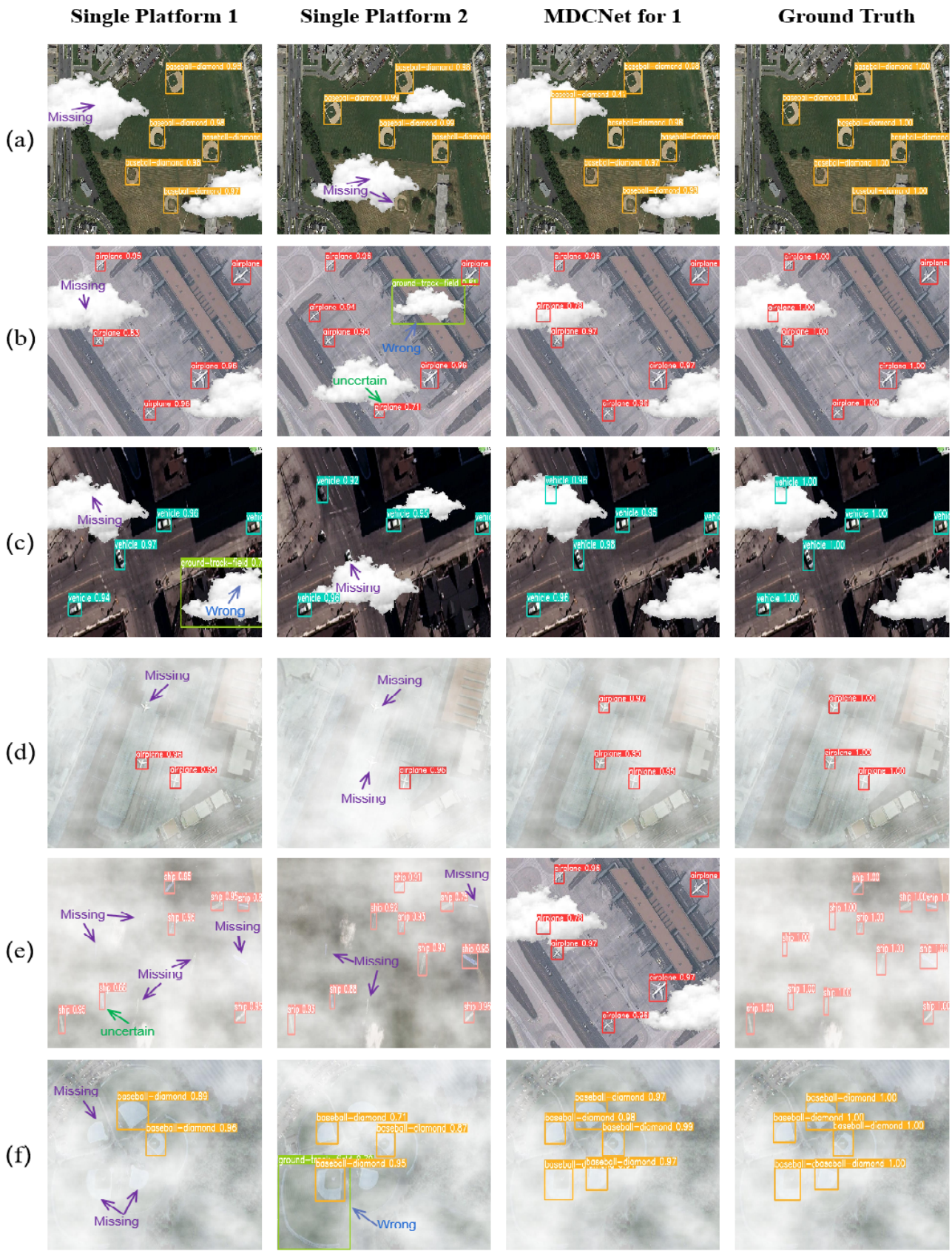


Fig. 10. Visual results of single platform and the proposed MDCNet.

objects under the cloud when the observation is disturbed, while after the multiplatform collaboration, it breaks through the observation field of the single platform and can warn the objects under the cloud. In addition, the proposed method can also accurately detect objects that are predicted incorrectly or with uncertainty due to interference, which further improves

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TABLE VII
EFFECT OF TRANSMISSION THRESHOLD

Transmission Threshold	NWPU-thick		NWPU-thin	
	$mAP_{50}(\Delta)$	comm ratio	$mAP_{50}(\Delta)$	comm ratio
No-Collaboration	68.0 (-)	-	45.0(-)	-
0(No-compress)	82.3 (↑ 14.3)	1.0/1.0	76.5 (↑ 31.5)	1.0/1.0
0.01	81.7 (↑ 13.7)	0.1624/1.0	75.3 (↑ 30.3)	0.1470/1.0
0.1	79.6 (↑ 11.6)	0.0560/1.0	72.1 (↑ 27.1)	0.0554/1.0
0.2	78.2 (↑ 10.2)	0.0235/1.0	71.7 (↑ 25.7)	0.0256/1.0
0.3	76.6 (↑ 8.6)	0.0120/1.0	70.7 (↑ 25.5)	0.0138/1.0
0.4	74.4 (↑ 6.4)	0.0027/1.0	70.5 (↑ 25.5)	0.0037/1.0
0.5	74.5 (↑ 6.5)	0.0007/1.0	70.5 (↑ 25.4)	0.0012/1.0
0.6	74.5 (↑ 6.5)	0.0001/1.0	70.4 (↑ 25.4)	0.0003/1.0
0.7	74.5 (↑ 6.5)	3.59e-5/1.0	70.4 (↑ 25.4)	7.08e-5/1.0
0.8	74.5 (↑ 6.5)	0.0/1.0	70.4 (↑ 25.4)	0.0/1.0
0.9	74.5 (↑ 6.5)	0.0/1.0	70.4 (↑ 25.4)	0.0/1.0
1.0	74.5 (↑ 6.5)	0.0/1.0	70.4 (↑ 25.4)	0.0/1.0

TABLE VIII
VALIDATION OF MULTIPLATFORM COLLABORATIVE NETWORK BASED ON FAIR1M

Source of Inference (SOI)		mAP_{50}	AV	Δ
Clear Image[UpperBound]		39.8	39.8	-
Single platform 1		25.0	24.95	-
Single platform 2		24.9		
MDCNet	platform 1	**28.2**	**28.05**	↑ 3.1
	platform 2	**27.9**		

the overall object detection accuracy compared to single platform.

V. CONCLUSION

In this article, we propose a multiplatform distributed collaborative inference framework in RS. MDCNet mainly considers the collaborative integration of observation information from multiple RS platforms. Compared with single-platform inference, under constrained observation conditions, MDCNet unites the observation information from multiple platforms, which supplements the insufficient information for a single platform and improves inference accuracy. Furthermore, considering the harsh interstellar transmission conditions, we design the SFC module to significantly reduce intermediate feature transmission costs by combining the object distribution characteristics in RS. After compression, only the key object feature regions are transmitted, which achieves a more favorable trade-off between low transmission cost and high accuracy gain. In addition, SCFF module provides spatial priors during the multiplatform feature fusion process, guiding more effective feature fusion to achieve higher accuracy.

We conduct extensive experiments as well as analysis on three public RS datasets, NWPU VHR-10, DOTA, and FAIR1M, and verified the superiority of the proposed method in terms of accuracy and communication efficiency. In the experimental scenarios, the collaborative strategy enhances the detection accuracy by 10.1%–13.7% compared with single platform, while compressing the interplatform transmission by more than 80%. However, the method still has limitations in the case of multiplatform asynchronous observations. In our subsequent research, we consider introducing historical time-series information to explore multiplatform asynchronous, incompletely aligned observation information integration methods, which aim to solve the problems of acquisition lag as well as inconsistent view angles. Besides, we intend to utilize multimodality data in RS collaborative perception to further achieve more generalized processing capabilities in distributed scenarios, which can be applied to practical scenarios with fast response and high accuracy requirements, such as disaster monitoring.

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